

Chapter 8

Input as a Critical Means to an End: Quantity and Quality of Experience in L2 Phonological Attainment

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Introduction

This chapter explores new understandings of the significance of input for long-term attainment in a second language (L2) with an emphasis on the phonological realm, where much of the relevant research is emerging. While the short-term impact of input has been given much attention recently (see Moyer, 2006a for a brief review of listening comprehension research), it is only over the long-term that we can see what lasting benefits input may have. I first contextualise the issue, outlining how input has been defined in the research traditionally, in order to contrast those early measures with new methodological developments. Age of onset and length of residence are briefly discussed, followed by more recent measures such as language contact and language use as indicators of authentic target language input.

Two main assumptions guide the discussion and conclusions: first, that *long-term attainment in L2 is a reflection of both quantity and quality of input*, and second, that *the learner's orientation to the target language is the main force behind how s/he utilises L2 input*. This focus on *input as experience* assumes a learner who is actively engaged in the acquisition process, not a passive bystander for whom exposure alone is sufficient – that is clearly not the case for most late learners, and even for many who begin early. The collective research points to the need for a greater focus on how and why input matters, vis-à-vis these indicators: (1) *sources of input*, namely, domains that serve personal and informal communicative functions; (2) *intention toward L2*, exemplified by the shift toward L2 as the primary language.

Traditional Measures of Input in L2 Phonology

Age of onset

Over the last three decades, second language acquisition (SLA) research has provided strong evidence for the negative impact of a late start in L2 (beyond early childhood), typically measured as AGE OF ARRIVAL to the TARGET LANGUAGE (TL) country (for immigrants). Here, I will use the term AGE OF ONSET, or AO, since it applies equally well to classroom and non-classroom learners (both in and beyond the TL environment). Brought to the forefront of SLA research by Johnson and Newport (1989) and Long (1990), the linear relationship between AO and attainment in morphology, syntax and phonology in L2 has been firmly established (Birdsong, 1992; Bongaerts *et al.*, 1997; Coppieters, 1987; Fathman, 1975; Flege *et al.*, 1995a, b, 1999; Jia *et al.*, 2002; Johnson & Newport, 1989; Moyer, 1999, 2004a, 2005, 2006a, 2007; Munro & Mann, 2005; Patkowski, 1980; Purcell & Suter, 1980; Thompson, 1991; see Hytlenstam & Abrahamsson, 2003 for a comprehensive review). At the same time, some SLA scholars dispute whether the CRITICAL PERIOD for language learning is truly 'critical', or unyielding, based on evidence for NATIVE-LIKE performance among small subsets of late learners (Birdsong, 1992; Birdsong & Molis, 2001; Ioup *et al.*, 1994; Olson & Samuels, 1982; Van Boxtel *et al.*, 2003; White & Genesee, 1996). The possibility of 'separate' critical periods for phonology versus other aspects of language seems strong, given that 'exceptional' learners are more prevalent when tasks are focused on grammar, as in grammaticality judgments (Birdsong, 1992; Birdsong & Molis, 2001; see Long, 2005 for a recent critical discussion), but appear far less often for phonology.

It is important to point out that evidence for and against a critical period in language learning has been problematic for a number of reasons. First and foremost, learners across studies are rarely matched in terms of experience with L2 (especially instruction) and motivation to sound native, meaning that the input received is not comparable, nor is learner orientation to the target language. For phonology, comparability has been particularly problematic because some pronunciation tasks are 'seeded' with difficult sounds (Moyer, 1999, 2004a, 2007; Thompson, 1991) while other studies avoid this in order to more readily fool the raters, thus offering a limited view of what constitutes phonological abilities (see Long, 2005 for discussion; also Moyer, 2006b). Conclusions from one study therefore do not necessarily (in) validate those of another, and scattered evidence for so-called 'exceptional' learners is too inconsistent to be generalisable.

The second, and perhaps most compelling, problem is that AO is inherently connected to other input concerns such as length of residence and instruction in the target language. This means that much of the purported evidence for a critical period has been over-interpreted, except where factors other than AO are taken into account.¹ A third problem is

that, despite a long history of evidence suggesting the importance of socio-psychological orientation factors (e.g. motivation, attitudes toward the target language, etc.), some scholars are dubious about the role they play, in part because an early start in L2 presents a kind of conundrum; it is typically associated with more positive affiliations and attitudes toward the target language, deeper connections culturally and socially, and so on. In other words, we cannot be sure which is the more direct or predominant force – age or AFFECT – for L2 learners; the observed significance of affective factors could be an ARTEFACT of their relationship to AO (see Oyama, 1976).² As a counterpoint, recent studies on learners across an AO range demonstrate the relative (statistical) power of affective and experiential factors for phonological fluency (Moyer, 2004a, 2007).³ At this stage, however, the directness of any given factor is unclear, and the potential conflation of age with its concomitant factors presents an ongoing methodological challenge.

These points of debate notwithstanding, statistical modeling should not be the predominant force guiding SLA research and theory. The gap in our understanding can be stated simply as this: While a late start could indicate (largely unobservable) neuro-cognitive constraints on one's ability to acquire language to a native level – and especially for accent – AO is essentially a reflection of exposure; it does not tell us anything about the structure or utilisation of the input received, for example, whether we can discern any patterns in the availability of authentic input for very young learners as opposed to older ones, or why some learners seek out more opportunities to use L2 than others. These considerations are important given that AO is highly correlated to instruction and education in the target language, contact with native speakers (NS), intention to reside in-country, desire to sound native, comfort with cultural and linguistic assimilation, and so forth (see Jia *et al.*, 2002; Moyer, 2004a, 2007). In short, because of its close association with numerous aspects of experience, input, and orientation, AO leaves much to be desired as an explanation for what is a very complex endeavour – one that is, by its nature, grounded in a social framework.

It is a truism that first language acquisition takes care of itself, meaning that all children become native-level speakers (assuming normal social, cognitive and psychological conditions), but this is clearly not so for those acquiring a language later in life.⁴ Here, authentic input and the opportunity to develop linguistic fluency in a supportive environment are no longer guaranteed, yet their importance does not diminish. These are fundamentally QUALITATIVE issues that have been treated only preliminarily in the research on age effects thus far. Keeping with the traditional focus on QUANTITATIVE measures of L2 experience then, we turn to length of residence, the second most common yardstick for measuring L2 input.

Length of residence

As a simple, continuous measure, length of residence (LOR) has long been studied as a factor in ultimate attainment in L2 phonology (Asher & Garcia, 1969; Flege *et al.*, 1995b; Moyer, 1999, 2004a, 2007; Oyama, 1976; Thompson, 1991; see also Flege, this volume). As with AO, its impact has been over-simplified (or under-explained) because, as a discrete measure only, it reveals nothing about the quality of TL experience. In addition, it is highly correlated to AO and thus its independent effect is unclear. It has been a mainstay in age effects research for at least two reasons, however: (1) it lends itself to quantification, and is therefore easily tested for statistical significance (but see Note 1); (2) in-country residence presumably measures the learner's exposure to authentic input, and thus, meaningful use of the target language. Anecdotally, however, we all know of immigrants with many years' residence in their adopted countries who never come close to native-like proficiency in accent (or morphosyntax). Perhaps because this second assumption is *not* universally valid (i.e. in-country residence does not guarantee quality input or interaction), statistical tests of its significance have been inconsistent (see Piske *et al.*, 2001 for discussion). Flege and Liu (2001) put this issue to the test by comparing 60 students and non-students (all Chinese immigrants to the United States) on phonemic recognition, listening comprehension and grammaticality judgements. The statistical analyses indicate that LOR is not significant for the non-students, meaning that extra years in-country do not matter. For the students, LOR is highly significant even when effects from other (possibly conflated) variables are partialled out. The authors speculate that the kind of input students received from native speakers, as well as teachers, made the difference, and this is why LOR was significant. In other words, just living in-country does not necessarily lead to greater attainment; input and practice play a pivotal role.

Like AO, LOR is not a terribly descriptive measure of L2 experience, and it is an unreliable predictor of L2 phonological attainment. Despite several decades of inquiry, there is little discussion of what underlies these statistical contradictions. One obvious conclusion is that mere exposure is not enough; late learners must engage in the L2 environment in certain ways, taking advantage of the surrounding input to further their fluency. Circumstances favorable to such endeavors naturally vary across learners, depending on age as well as ethnic, social, educational, and professional background. It is no real surprise, then, that different populations in different settings produce variable results. A recent example illustrates the complexity underlying the LOR construct. In a recent study of immigrants to Germany with extensive experience in the language, Moyer (2004a) was able to correlate performance on a number of tasks (read-aloud and extemporaneous, guided speaking) with specific L2 input factors such as contact

with native speakers, instruction in the target language, and many others. Measurements of these factors included quantitative survey data on the numbers of hours per week, as well as qualitative survey and interview data on the consistency of contact with native speakers over time, contexts for L2 use, sense of self and identity in the second language, and so on. An analysis of introspective, guided interviews for these learners (AO mean 12 years; LOR mean six years) supported the conclusions from the quantitative analysis. Together, this integrated approach to the experience issue confirms the following psychological, social, and cognitive implications of LOR:

- (1) *Psychological*: LOR correlates to a sense of satisfaction with attainment, personal motivation toward L2, and sense of overall fluency.
- (2) *Social*: LOR correlates to frequency of spoken contact with NS, and intention to reside permanently in-country.
- (3) *Cognitive*: LOR correlates to overall years of L2 instruction as well as L2 educational experience (other subjects taught in L2), instructional emphasis on interactive and communicative use of L2 (not just formal translation-type activities), amount of classroom feedback on pronunciation, and types of phonological training (Moyer, 2004a: 143).

In sum, a longer LOR indicates optimal instruction and targeted feedback, as well as greater contact with native speakers, and implies numerous other social and psychological benefits. Importantly, similar relationships are found for AO among this group, with one additional noteworthy set of correlations: Older learners report greater reluctance to initiate social contact with Germans, and do not typically ask for feedback on their German, that is, they do not seek to improve their fluency in the same ways as younger learners do. Taking all of this into consideration, we can conclude that learner orientation and input work together as they decisively impact attainment; those with more instruction and more personal contacts are judged significantly closer-to-native in accent, and they express greater sense of confidence and identity in L2 (see Moyer, 1999, 2007 for similar findings).

Not surprisingly, AO and LOR typically demonstrate overlapping connections to these kinds of experiential and socio-psychological factors (see Jia *et al.*, 2002; Moyer, 2007; Purcell & Suter, 1980). Yet because such connections are seldom acknowledged as relevant to the critical period/maturation issue *per se*, we find ourselves at a sort of methodological standstill. Many facets of L2 experience and orientation underlie traditional exposure-type measures like AO and LOR. Only by examining how these concomitant influences are interwoven can we come to a deeper understanding of their observed impact.

Recent Research: Input as Language Contact and Use

With the prominence of LOR and AO firmly established in the research, several detailed aspects of L2 input have recently come to light, especially for the phonological realm. This section examines how language use and contact reflect not just exposure, but the actual utilisation of input to positive ends.

One way to measure input beyond years of total exposure is to look at language contact, or opportunities for language use. This is not so straightforward, however, and has rarely been approached empirically for that reason. Language contact, in general, can imply any number of things, including formal instruction, structured or unstructured activities beyond the classroom, personal communications, daily business transactions, academic or work-related endeavours, and so on. This implies a range of INPUT QUALITY, authenticity, formality, interactivity, and so on, and thus, specific criteria are needed to narrow down some points of focus. This section briefly considers both the quantification and the qualification of L2 contact and use, based on a few specific criteria recently shown to be significant for L2 phonology.

Time on task

Perhaps the most common measure of language contact among instructed learners is the number of semesters or years spent studying the language. It is highly significant for long-term outcomes in L2 phonology according to a number of studies (Diaz-Campos, 2004; Elliott, 1997; Flege & Liu, 2001; Flege *et al.*, 1999; Moyer, 1999, 2004a; Purcell & Suter, 1980), but not consistently so (*cf.* Flege *et al.*, 1995b; Thompson, 1991; see Piske *et al.*, 2001 for review), perhaps due to the fact that it is not as straightforward a measure as it appears. There are inevitable differences in the quality of instruction, both because of the teacher's own linguistic skills, and the inherent circumstantial differences of setting, curriculum, pedagogical approach, class size, and so on. This makes comparisons across studies somewhat tenuous.

As for what kinds of instruction are most effective for developing a more authentic accent, communicative approaches appear to be preferable to grammar-translation (Moyer, 2004a), likely because they focus on meaning and interaction, and thus assist in the possible restructuring of L2 hypotheses and learner 'uptake' of new target forms (Lyster & Ranta, 1997; Swain, 1995; Swain & Lapkin, 1995; see Pica, 1994 for review). Overt instruction on suprasegmentals like stress, pitch and intonation patterns also appears to be significant, though too few studies have looked at this factor, so these are preliminary conclusions only (see Derwing & Rossiter, 2003; Elliott, 1997; Missaglia, 1999; Moyer, 1999, 2004a). At this point, we

can (cautiously) conclude that instruction does represent input and practice, at the very least, and that exposure to it is beneficial for accent, especially when activities are targeted to segmental and suprasegmental accuracy and authenticity.

Another way to test language contact is to measure the number of hours spent using L2 beyond the classroom. With regards to hours on task, two criteria have recently demonstrated significance for phonological attainment: (1) weekly or daily hours spent on L2-oriented activities; (2) use of L2 relative to L1. Simply looking at hours on task, one of the most interesting sets of findings comes from an early study by Purcell and Suter (1980) which shows that contact with native speakers in formal (work and school) domains is significant for accent, but not nearly as significant informal contact, such as living with a native speaker (measured as total contact months) (see also Flege & Fletcher, 1992; Tahta *et al.*, 1981). More recently, segmental accuracy has been shown to improve based on amount (days per week) of L2 use beyond the classroom (Diaz-Campos, 2004). (Syntactic performance has similarly been shown to benefit – see Moyer, 2005 for supporting evidence.) As for L2 use relative to L1, Flege and his colleagues have established strong support for this criterion (Flege & MacKay, 2004; Flege *et al.*, 1995b, 1999, 2002; *cf.* Flege & Liu, 2001; Thompson, 1991). To (very) briefly summarise this relationship: the more L2 use, the better, for both production and perception of non-native phones and phonemes.

To summarise, time on task does have benefits for second language accent – understanding that impact will vary across individuals – but the evidence is not terribly consistent. Any conclusions about how much is ‘enough’ would thus be premature at this point. Having examined *how often* or *how much* input makes a difference, we turn to the underlying – and more vexing – question: *must input be directed, or used, in certain ways?*

Contexts for L2 use

Once we move beyond easily quantifiable measures, the boundaries become fuzzier and the evidence far more scarce for *qualifying* L2 use.⁵ Although sociolinguistics has long confirmed compelling evidence for contextual patterns of language variation (see Giglioli, 1972) there is noticeable resistance to this paradigm within the cognitively-oriented mainstream of SLA research. The past several decades have seen a growing division between cognitively-oriented SLA scholars, and those who support a more integrated, externally-focused framework (see Collentine & Freed, 2004 for discussion). For our purposes here – examining the critical importance of input – straddling these paradigms is unavoidable; input is essential to any cognitive processing model because input fuels knowledge restructuring. It also underlies any sociologically – or psychologically – constructed model because receiving input implies communication within

a learning environment, and this naturally calls to question all sorts of social and psychological issues that impact input processing. Bringing these points to bear on the language use discussion here, two specific criteria have recently come to the fore as significant: (1) the use of L2 in informal, personal domains; (2) the multiplicity of L2 contact domains, discussed below.

Using the target language informally, especially to build personal, social connections beyond a formal instructional setting, is clearly significant for long-term syntactic (Moyer, 2004b, 2005), phonological (Flege *et al.*, 1995b; Moyer, 2004a; Purcell & Suter, 1980), and even listening comprehension abilities (Moyer, 2006a). This can be determined by surveying both frequency and consistency of contact, measured as hours of contact with native speakers per week (Moyer, 1999, 2004a, 2005, 2006a, 2007). In addition, participants may describe their 'primary source of target language use' as either native speakers, other non-natives, or a combination. In Moyer's studies on accent, accent is consistently correlated to primary contact with native speakers while in-country. In addition, interactive, conversational kinds of contact and language use, as opposed to passive or receptive activities that involve mostly listening or writing, are far more likely to correlate to native-like performance (Moyer, 2004, 2006a, 2007). As for access to multiple contact domains, the combination of instructional (formal) *and* informal contexts is a far more powerful predictor of attainment than is either type by itself, for several aspects of L2 ability (Moyer, 2004, 2005, 2006a, 2007).

While the exact nature of these connections is unclear, it is fair to conclude that the interactive, personal nature of the input in these contexts is key. Several findings support this conclusion. First, greater informal contact with native speakers is highly significant for accent (Moyer, 2004a, 2007; see also Flege *et al.*, 1995b). Second, living with a host family is apparently more beneficial to attainment than is a dormitory-type experience for several types of language fluency (Moyer, 2005), possibly because a family setting ensures a consistent, intimate setting for interactive language contact (also, a dormitory setting does not ensure contact to native speakers). Simply put, a home environment is a sure bet for optimal input and practice. Moreover, living with a family likely enhances one's perceived need (and desire) for greater fluency because meaningful communication is a necessary precondition for comfort in that environment.

To summarise, recent work shows that L2 use positively impacts several levels of language fluency, including phonology, when it is characterised by a variety of contact domains. The extent of native speaker (NS) contact is not just about hours of use; it represents meaningful opportunities to take in (and generate) authentic input. This makes sense from a cognitive processing point of view because multiple and complex opportunities for L2 use ensure rich input and meaningful communication (which is

undoubtedly circular if more contact leads to more language use). From a socio-psychological point of view, multiple contact domains (formal and informal, interactive and receptive) indicate that various communicative and social functions are being served (for supporting evidence, see Moyer, 2007). Here we are reminded that language fluency develops in conjunction with language affiliation and identity: Multiple opportunities for language use indicate a deep investment in L2, all of which leads to growing confidence and ability (also circular).⁶ For phonology specifically, several researchers have examined the significance of identity, investment, and sense of self or place in the socio-cultural context, that is, against the backdrop of the target language community (Major, 1993; Moyer, 2004a; Piller, 2002).

While these orientation factors are associated with cognitive and social strategies for improving fluency (e.g. consciously imitating native speakers' production of difficult phonemic contrasts, joining organisations to enhance NS contact, avoiding L1 speakers in favour of L2 use, as shown in Moyer, 2004a), it is not easy to understand the role these factors play through discrete measures alone. Qualitative data can help point to fuller and more appropriate interpretations of quantitative data (such as that provided by scalar and categorical questionnaire responses). Moyer's (2004a) integrated analysis of quantitative and qualitative data for immigrants to Germany shows that greater confidence in L2 and contact with native speakers are closely tied to strong affiliation with the target language, both linguistically and culturally. The combined analyses clearly show that identity relates to assimilation as well as to attainment. Furthermore, a closer analysis of several participants' interviews shows how complex, and even contradictory, the identity issue can be. Some learners speak openly about their conflicting senses of affiliation between L1 and L2 (culturally as well as linguistically), and several also separate investment in the language from a cultural affiliation when political, cultural, and social orientations of the host country present a moral or cultural dilemma, or result in negative experiences (such as discrimination).

Because these socio-psychological issues are so complex, and because they shift in response to changing circumstances, it is not possible to confidently pinpoint their impact on attainment. It is likely, however, that these factors lead to specific behaviors, which can be directly identified and measured with some degree of accuracy and predictability.

Connecting Input and Attainment in the Big Picture: Quantity and Quality Concerns

The discussion above supports the idea that input should be more directly viewed as a reflection of language contact and use, especially in reference to learner orientation, since this determines how the learner

actually *utilises* the input available. Simply put, successful attainment in a second language – in the first language as well – relies on optimal levels of input, quantitatively and qualitatively.

As discussed above, quantity of input has been operationalized as length of residence, years of instruction, and more recently as time on task and proportion of L2 to L1 use. All appear to play a role in attainment, and the evidence cited here generally supports the significance of overall amount of input, though time on task does not tell the whole story. For one thing, amount of input is not (universally) sufficient to push the learner toward native-levels of ability, as evidenced by late learners who have lived years – even decades – in-country, yet maintain a distinctly non-native accent. Quantity may therefore be a better predictor of attainment if examined in terms of actual L2 use, but to truly capture how this makes a difference, we must find out *why* learners use the target language as they do, that is, for what functions and to what social and psychological effects. This leads us to the importance of input *quality*, which concerns the context-bound conditions for L2 use. The evidence cited above clearly shows how fundamental input is to attainment in a second language accent, summarised by these conclusions:

- (1) Contexts for L2 use correlate significantly to accent (and morpho-syntax) when they include both personal/informal domains – not just formal/professional ones – and when they are available from a number of sources.
- (2) Consistent contact with native speakers – beyond brief business-like transactions such as standing in line at the grocery, ordering food, and so on – is significant for transcending predicted age constraints on phonological acquisition.

Another way of stating this is that rich opportunities for language use underscore what is essential for native-like attainment in phonology: authenticity and breadth of meaningful practice. Using L2 across multiple domains – and with native speakers – inherently means that many communicative functions are covered, including: expressive, emotional and social ones, as well as referential, or concrete ones, like exchanging basic information (the kind of function that language instruction is typically limited to). Attainment is clearly best served by rich and varied language use opportunities. One additional conclusion – more preliminary, perhaps – is that the structuring of language use opportunities is a purposeful endeavour, and learners are more likely to pursue them when they feel no threat, either socially or psychologically. Recent qualitative methods have opened up the research along these lines so that can we explore the essence of how learners and input connect.

Acknowledging that SLA is effortful and complex – not something that takes place in a laboratory setting, or by default when one resides

in-country – the evidence cited here underscores the necessity, and consequences, of learner engagement. For phonology in particular, recent work indicates that experience and orientation constitute a kind of circularity: Those with greater opportunities for L2 use also express more positive orientations toward the target language and culture, and develop social and cognitive strategies to increase language contact (Moyer, 2004a, b), which, in turn, increases meaningful input and practice. To illustrate the consequences of this circularity, I refer again to the German context. Many immigrant communities in Germany face poor living conditions and relative social and geographical isolation (see Fennell, 1997; Moyer, 2005 for a review of early studies on immigrants to Germany, including the ZISA⁷ database; cf. Schumann, 1978 for his well-known study on ‘Alberto’, a Spanish speaker acquiring English in the United States). To what extent can L2 German users develop real linguistic fluency under such conditions, especially with limited (or no) access to formal instruction, few possibilities for establishing personal contacts with Germans, and no intention of residing permanently? With this in mind, the quality/quantity issue can be boiled down to one simple and obvious maxim: *The opportunity to use language meaningfully across various contexts, and serving multiple communicative and social functions, is fundamental to native-like attainment, whether in L1, L2 or beyond.*

Conclusions: Moving the Research Forward

Throughout this chapter, I have emphasised the importance of social and psychological orientations, particularly for phonology, where age constraints appear most prevalent, and where identity can constitute a place of resistance. In the interest of maintaining their established identity, some late learners are reluctant to pursue a native-like accent in L2 (Moyer, 1999), even when they do not live in the target language environment. Where SLA takes place as a byproduct of immigration, identity concerns can be more acute, especially when the extant social and political tenor determine the L2 user’s potential to assimilate and/or integrate. If quality and quantity of input are limited, and expectations (or potential) for assimilation are low, attainment predictably suffers. Looking at it from the individual’s point of view, if the target language is not deemed to be critical, neither to survival nor to self-image, there is little incentive to pursue advanced fluency.

Along with the individual’s own set of motivations and orientations, SLA is subject to broader realities, like family attitudes and support for L2, access to formal instruction and/or education in L2, and so forth. These conditions should be acknowledged as ‘gatekeepers’ to input, and therefore as powerful influences on attainment. Understanding the individual’s circumstances within these broader frameworks requires us to expand our

methods to incorporate multiple types of data, especially qualitative data to augment quantitative analysis.

Coming back to the overarching framework of this chapter, if AO is in fact the most reliable indicator of attainment – especially for accent – can we explain its impact in terms of L2 input and use? The statistical link between age and outcome has long since been established. What is needed now is a greater understanding of what maturation implies in terms of the process of acquisition, in other words, what age means for *need*, *desire* and *opportunity to become native-like*.

Based on evidence from many studies, I have reiterated here (and elsewhere) an obvious generalization for SLA, namely, that an early start predicts a greater variety of input sources, ranging from formal to informal, and covering a wide array of contact domains. The resulting opportunities to use L2 lead to greater confidence and to greater fluency. To illustrate these inherent interconnections between learner and input schematically, Figure 8.1 below brings together age of onset with its primary co-varying influences, shown here as ‘clusters’ of factors (i.e. multiple, related influences):

- (1) the cognitive processing cluster (instruction and feedback received, learner strategies, and so on);
- (2) the social processing cluster (contact with NS, language use domains, and so on);
- (3) the psychological processing cluster (reasons for learning L2, attitudes, and so forth), as they relate to the learner’s access to, and utilisation of, authentic input in the target language. The primary realms of L2 processing are highlighted in bold, with the fundamental connections between them in the smaller (italicised) boxes (e.g. access and opportunity, styles and strategies, and so on).

In this figure, input takes its place in the center, as each of these processing clusters – along with AO – directly affect how the individual learner utilises input. The term ‘strategic use of input’ here emphasises the fact that learners make choices, taking advantage of available input for language restructuring purposes in accordance with their own cognitive styles; setting specific intentions, goals and outward behaviours to suit their psychological perspectives; constrained (or perhaps encouraged!) by avenues of access and opportunities to develop greater contacts according to social orientations, and so on.

The objective of such a schematic is not to oversimplify the conundrum that is SLA; it is to underscore the integration of these multiple realms of influence and to highlight the (idiosyncratic) nature of what connects them. It must be noted that these four main influences (AO along with social, psychological, and cognitive factors) are presented here as universally

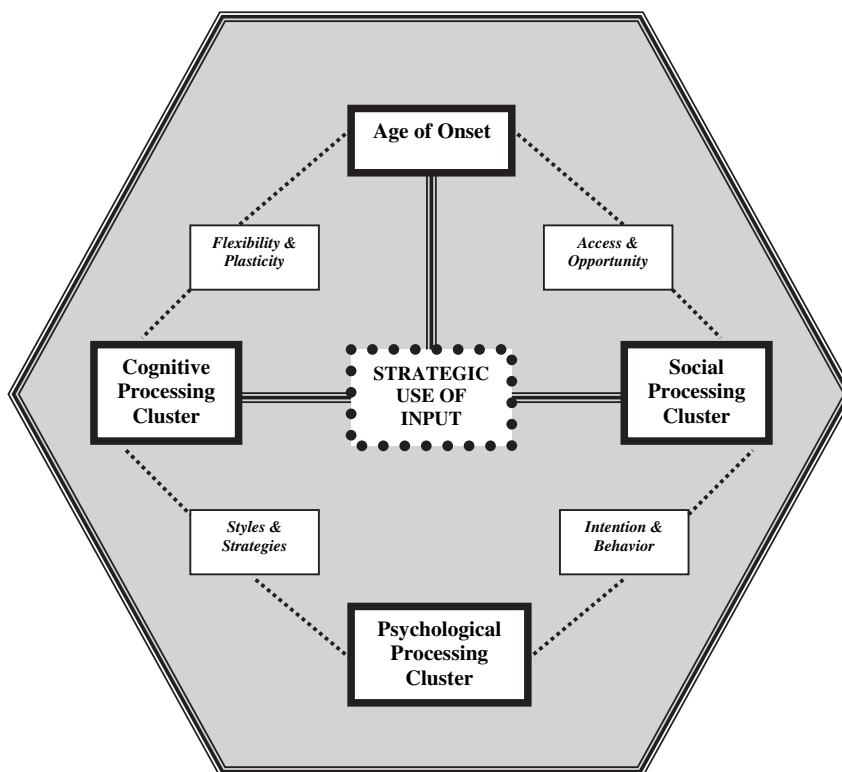


Figure 8.1 An Integrated view of critical influences in SLA (adapted from Moyer, 2004a)

relevant – presumed critical to the processes and outcomes of SLA – however, the balance of their contribution varies according to the individual learning situation.

To really push the research forward, we are at the stage where greater exploration is needed, not just confirmation of the same tried-and-true relationships. The problem lies in figuring out where to go from here. Recognising the circularity between many influential factors in SLA, can we ever know which factor, if any, takes precedence? Maybe not, but we can reframe our focus to explore what kinds of input are (typically) available to learners of specific ages, and how these language contact sources contribute differentially to attainment. Such a focus on input as a critical means to an end would shift the prevalent emphasis away from ‘failure’ among late learners, and instead appreciate the cognitive and social flexibility required under (often) difficult circumstances.

Finally, considering the research over the past several decades as an evolving whole, I put forth two considerations that, to my mind, merit a great deal more attention as possible focal points for the study of input:

- (1) The first area could be called sources, or domains, of input, and refers to the inclusion of personal, informal communication within one's repertoire of language use domains. By personal spheres of contact, I do not mean simply greeting one's favorite shopkeeper once a week, but engaging in real communication where some deeper social or emotional function is served, such as establishing a new friendship. To the best of my knowledge, it is not possible to attain native-like fluency – at least not for phonology – without crossing over from the formal to the informal and personal mode, and only then if consistency in that sphere is maintained. Those late learners who make personal connections, and maintain these important sources of L2 practice over time, are most likely to attain a native-like command of the phonology, and other aspects of language as well. This is possibly due to the fact that communicative interaction inherently involves a 'negotiation of meaning' (see Long, 1996), which could benefit long-term attainment by aiding learner comprehension of new linguistic features (Pica *et al.*, 1986; Swain & Lapkin, 1995), possibly leading to 'uptake', or what learners attempt to do with the feedback received (Lyster & Ranta, 1997). In other words, meaningful negotiation is likely far more significant for language acquisition than are so-called 'passive' exposure activities (see also Gass & Selinker, 2001; see Moyer, 2007 evidence regarding phonological attainment).
- (2) The second consideration I have called *intention toward L2* for lack of a more precise term (Moyer, 2004a), but in the context of this focus on input, one aspect seems particularly revelatory: the shift in language dominance toward L2. Thompson (1991) has suggested that those who lose their mother tongue stand a significantly higher chance of sounding native in L2 than those who maintain it (sad to say). Singleton (2000) makes a similar point. He argues that an AO of 10 years or earlier is indicative of a shift to L2, such that we are no longer really looking at 'L2' anymore if the target language has supplanted the mother tongue. This chapter has noted several interesting findings that support the veracity of these observations. The L2 user's access to multiple contact domains, a long residence [at least six years, according to Moyer's (2004a) data, and as many as 13 or more, according to Moyer (2007)], and the intention to reside long-term, are all significant to attainment – and all overlap with one another (and not coincidentally, with AO as well). This combination suggests a learner profile of sorts in that this trio of experiential and orientation factors predicts a fundamental shift in language affiliation. Such an assertion

holds up in light of the (inconclusive) results for traditional measures such as LOR: Where LOR results are robust, we are seeing evidence for learners who have made this fundamental linguistic shift. Since many learners never do, variable results for LOR are prominent. In other words, where participants across a broad range of LOR are studied, we find great variability because we capture those 'in-between' learners who are in the process of either shifting their language affiliation or holding firmly on to L1. When a clear dichotomy among study participants exists (e.g. Group 1 has 10+ years' LOR compared with Group 2 at 2 years or less, as in some of the research by Flege and colleagues), the outcomes are far more powerful for LOR. In the rare instance that LOR is further analysed for contact with NS and L2 use, its effects appear to be (statistically) independent from AO (such as in Flege *et al.*, 1999). This is a critical piece of evidence for my assertion here, because once that transition in language affiliation is made, the relationship between accent and LOR becomes clear and predictable. To my knowledge, this particular angle on the language dominance issue has only recently been addressed for morphosyntax (see Jia & Aaronson, 1999, 2003), but has not been investigated as such for phonology.

For purposes of validation, we first need more exploration of L2 experience and input through qualitative data to develop a reliable set of measures that can be replicated. A solid beginning could be established through introspective reports and interviews – in conjunction with linguistic measures, of course – preferably on a longitudinal basis in order to capture how these language affiliations respond to external circumstance and socio-psychological orientation – neither of which is a static source of influence.

A gradual, yet substantial, shift is evident in the research in SLA, with L2 phonology front and center in this new emphasis on input and language use. Importantly, L2 learners are no longer viewed as 'deterministic input-output machines' (Jia & Aaronson, 2003: 133), as they once were. They are active participants in the process, even helping to construct the input, seeking out ways to make the best of whatever limits they perceive, both externally and internally imposed. Revising traditional models of inquiry to account for these complexities will help move the research forward to greater understandings of why input is so important.

Notes

1. ANOVA can verify the independence of certain factors, and multiple regression analyses can check for the predictive power of confounded factors against one another, for example, AO versus instruction or LOR (see Piske *et al.*, 2001 for discussion; see Flege & Liu, 2001; Moyer 1999, 2004a, 2007 for this kind of analysis).

2. Moyer (2004b) points out that some studies supporting the significance of sociocultural and sociopsychological factors lack actual linguistic data for the purposes of correlation. Some do include self-ratings of L2 abilities, but these are less reliable than measures like achievement tests, native speaker ratings, and so on. A balance of linguistic and non-linguistic factors is essential if we hope to capture the relative contribution of input factors in SLA (Moyer, 2004b: 52).
3. The combination of a survey and introspective interview data in Moyer (2004a), allowed for a deeper analysis of the acculturation process, the role of L1 and L2 for the individual's sense of place in the target language community, and the importance of linguistic and cultural resources for maintaining ties to the heritage language and culture (as well as acquiring greater skill in the second language). For specifics of the instrument and the analyses, see individual studies cited.
4. An early start may be optimal, but it does not ensure native-like attainment (see Flege *et al.*, 1995b; Moyer, 2007; Thompson, 1991). In the context of SLA research, NATIVE-LIKE is typically determined to be performance that matches the performance of native speakers within the same study, serving as a 'baseline' for the production of target sounds. NATIVE SPEAKERS are generally understood to be users of a language with exposure to that language since birth or very early childhood, having used it consistently throughout their lives. As noted here, some research shows that a very early start is not sufficient to assure such a 'target' match (cited above), thus, concepts like language dominance may be a more appropriate consideration (depending on the individual's circumstances). In short, the construct of 'nativeness' is not as black and white as it may seem. In fact, it has been criticised somewhat rigorously in recent scholarship for upholding an ideal that is both (unrealistically) monolingual and narrow in scope (see Cook, 1999, for example).
5. See Flege and MacKay (2004) and Flege *et al.* (1999) for the impact of language use in terms of L1 usage domains. The relevance here is that frequency of L1 use – specifically, maintaining L1 across multiple and personal contact domains – is detrimental to the ability to produce and perceive L2 phonemic contrasts accurately. The authors maintain that this is due to the continued development of the L1 phonetic system, which has an impact on the perception (and development) of L2 phonetic categories.
6. See MacIntyre *et al.* (1998, 2003) for the 'willingness to communicate' (WTC) model as it relates to these notions. See Gardner and Lambert (1972) for early work on motivation in SLA.
7. ZISA is the acronym for the database, *Zweitspracherwerb Italienischer und Spanischer Arbeiter*, which resulted in numerous publications by H. Clahsen, J. Meisel, M. Pienemann, P. Muysken, and N. Dittmar, among others, from the 1970s through the 1980s. It has inspired a great deal of interest in uninstructed learners, with data generally supporting the idea of a 'universal' order of acquisition for certain syntactic and morphological features in German [but see Moyer (2005) for critical review; see also scholars such as Pfaff (1992) and Vainikka & Young-Scholten (1996, 1998) for more functional and input-based accounts of acquisitional phenomena in L2 German].